

Animal Narratives and Emotional Resonance in #ClimateChange Discourse on Social Media: A Qualitative Content Analysis

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ABSTRACT

Teens, who are online almost constantly, engage with climate change posts on social media. We present the findings of a qualitative content analysis of the most popular climate change posts (tagged with #ClimateChange) on Instagram, TikTok, and Twitter. Over four weeks in Spring 2022, we collected the top five weekly posts ($n = 60$) to understand how they address animals and emotions and relate to teens' perceptions of climate change online based on prior survey work with 100 American teens. Our research reveals that more than a quarter (26.7%) of all analyzed posts relate to animals, and many (87.5%) employ negative language when discussing animals. We offer recommendations to enhance online climate change communication for teens, foster pro-environmental behaviors, attitudes, and activism, and study social media discourse across platforms. We highlight the need for further research on youth, climate change, and social media.

CCS CONCEPTS

• **Human-centered computing** → Social media; Empirical studies in collaborative and social computing; • **Social and professional topics** → Adolescents.

KEYWORDS

Teenagers, Climate change, Social computing, Youth

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1 INTRODUCTION

Climate change is a universal crisis [15] and is often a part of online and offline discourse [33]. As climate change-related disasters increase in frequency and severity [12], it is crucial for HCI researchers to consider how their designs and studies can help mitigate this challenge. In our paper and larger research agenda,

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we focus on teen experiences with climate change on social media. Users can learn, discuss, and think critically about climate change online [2]. Social media provides a space to mobilize activists [2, 37], raise public awareness [23], highlight news [31], and discuss environmental politics [5, 42]. Social media assists different populations in engaging with online climate change movements, including teens [7, 47]. From social media, teens can more easily communicate with others [21], build their communities and engage in informal learning [34], and hear about current events [29]. Teens are highly engaged in social media [3, 45] and are more concerned about climate change than adults, as shown by higher levels of discussion about climate change online and offline compared to other age groups [43]. Research shows that emotion is an important element of engaging with climate change information [9, 39, 46], with young people's experiences functioning differently than adults' [30]. Further, discussion of animals prompts high levels of emotional engagement [32, 39]. Presently, there exists a lack of understanding regarding the emotional impact of climate change-related social media content pertaining to animals and across multiple platforms. Further, research on teens' experiences with climate change on social media is scarce.

To address these gaps, we report on a portion of a detailed qualitative content analysis of the most popular climate change posts (tagged with #ClimateChange) across three social media platforms (Instagram, TikTok, and Twitter). We collected the top five posts each week for four weeks ($n = 60$) in April and May 2022. We focus on how these posts address animals and emotion and relate this to teens' perceptions of climate change online, drawing from previous survey work with 100 American teens. **We ask: What kinds of emotions are prompted at the intersection of animals and climate change on different social media platforms, and what considerations are relevant for teens?** We find that 26.7% of all posts were animal-related and that the majority of these (87.5%) use negative framing to discuss animals, typically with emotional, vague calls to action. We contribute (1) insights on how to better communicate climate change information online for teens, (2) implications and open opportunities for attitude/behavior change and activism, and (3) discussion on the differences between #ClimateChange discourse on Instagram, TikTok, and Twitter. We conclude by describing the potential for a larger research agenda centered on youth, climate change, and social media. We hope to promote pro-environmental attitudes and behaviors in young people by encouraging their engagement and emotions toward this issue.

2 RELATED WORK

Climate change communication can inform behaviors and attitudes [32, 48], and we are interested in helping young people to develop

pro-environmental orientations. Social media is a valuable arena for this work, as evidenced by exemplars from literature on how social media has the potential to affect teen attitudes and behaviors [20, 35]. To do that effectively, we need to understand youth experiences with climate change online and characterize the larger social media climate change information and communication landscape. Few studies cover teen perspectives on climate change online, and literature tends to focus on individual social media platforms, such as Twitter [7] or TikTok [6]. Studies that cover climate change on social media (such as those using the #ClimateChange hashtag) are quantitative [49], computational [25], or focus more on engagement than specific post content [6], again targeting individual platforms. There is a scarcity of information on how climate change is depicted on multiple social media platforms, and few HCI researchers have studied this space.

Emotions play a key role in encouraging individuals to become involved in climate-related issues and take action [9, 28, 46, 48]. Emotions heavily influence people's reactions to climate science and its solutions, and climate communication that evokes emotions can counteract biases and engage audiences who avoid the topic [48]. Both affect and emotion are key drivers of climate change perception and action, and communications that evoke positive and negative emotions can encourage sustainable behavior [9]. Research has found that individuals with stronger positive emotional responses to climate change are more concerned about and feel closer to the issue [48]. Ignoring the emotional impact of climate-related news can decrease communication and outreach effectiveness [26], and researchers argue that climate messages intended for children should evoke positive emotions instead of guilt [30], as the latter may have negative consequences [19, 27].

Young people are worried about animal and plant extinction [38] and generally view harm to animals as more severe than damage to plants [14]. When teaching youth about climate change, focusing on animals may be an effective way to motivate environmentally-friendly actions [32]. Considering social media engagement, previous HCI researchers have used emotions as a lens for understanding topics such as depression [13, 44], postpartum changes [16], and cyberbullying [4, 11]. We extend this by considering how emotion connects to climate change and pro-environmental attitudes and behaviors.

3 METHODOLOGY

Emotions play a crucial role in individuals' attitudes toward climate change information. Social media platforms have rich multimedia content, including text, images, and videos. Building on a previous exploratory survey, we aim to better understand the emotional components of animal-related climate change content with qualitative content analysis [18]. Compared to more quantitative methods, our approach is well-suited for deep engagement with these kinds of diverse and detailed data, as well as our research question, which aims to characterize emotions related to social media posts.

Exploratory survey and data analysis. In Spring 2021, we conducted an exploratory online survey with 100 American teen participants recruited via convenience sampling. The ethics board-approved survey was created to learn about teen experiences with

climate change online. We employed thematic analysis [8] to analyze open-ended survey responses, utilizing an open, inductive, and iterative approach. Check-in conversations throughout the coding process were modeled on intercoder agreement checks [17]. From this survey, we realized a need to understand climate change discourse on different social media platforms, as teens had varying experiences with it depending on the platform. Additionally, some noted that animal-related posts were the most impactful and easily memorable, sparking a deeper research interest for us.

Qualitative content analysis. Our interests also expanded to better understand climate change content on different social media platforms. In Spring 2022 (April and May), we used the #ClimateChange hashtag on Instagram, TikTok, and Twitter, conducting our qualitative content analysis [18] to learn and compare how this crisis is spotlighted on different social media platforms. These platforms vary in the ways media are formatted in posts. Text, audio, and video are used in different ways and with different frequencies. We compared discourse across platforms to consider how different communities approach climate change. We were also motivated by the fact that the teens we surveyed experienced climate change discourse on multiple platforms. We collected data over a month (five posts a week per platform for a total of 60 posts). Earth Day, an environmental celebration and a day of awareness-building, occurred during data collection and was discussed in some analyzed posts. Based on Schreier [36], we created new accounts on all platforms to gather data and analyzed the top (most popular) posts for each week. To do this, when searching #ClimateChange, we used the "Top" tab on each platform to collect popular posts based on engagement for each week to better understand broad discourse around climate change rather than aiming to identify teen-created posts. We then qualitatively analyzed the top five comments for each post (the comments that had the most likes) to learn more about the discourse around the posts. To protect commenter identities, we used ethical fabrication [22] to make slight changes to featured comment quotes. We each independently coded each post and its associated comments, conversing with each other throughout the coding process to update and edit codes [17]. Through an iterative process, we identified a significant theme of animals and animal representation. All of these posts were also coded for patterns of strong emotion, aligning with what was reported in our survey.

4 FINDINGS

We found that 16 out of 60 (26.7%) social media posts were associated with animals, and we focus here on these posts because this was such an important theme across our data. We acknowledge that this is a small sample but believe future work can build on these preliminary exploratory findings. These social media posts often had multiple types of animals included. Just two posts contained only text. The majority of climate change posts concerning animals were seen on TikTok, while the least was on Twitter. Below, we discuss emotions and conversations about climate change in the context of these posts. We relate what we saw on social media to participant responses from our survey, highlighting the role of climate change information online and teen experiences with it.

4.1 Fatalism, Pessimism, and Shame

Social media posts that address the impact of climate change on animals are often imbued with strong negative emotions, emphasizing sorrowful sentiments. Out of the 16 animal-based climate change posts on social media, 14 of the posts (87.5%) framed animals as being harmed or negatively affected due to the changing climate, creating gloomy feelings. Drawing on TenHouten's concept of emotions of powerlessness [41], we argue that some posts and their associated comments may infer feelings of fatalism (powerlessness and feeling that control is in external forces and powerful others), pessimism (expecting things not to turn out well), or shame (sadness and fear). These posts relate to the survey we conducted with teens, as climate fatalism, pessimism, and shame were prevalent in the responses. We note that while these negative posts may encourage environmental action, they frequently lack specific guidance on meaningful steps.

Fatalism. Explicit, concrete examples of animal harm due to climate change may lead to climate fatalism [24]. We discovered that social media posts could demonstrate climate fatalism, even if it is not mentioned directly in the post. We encountered three posts where fatalistic comments were made despite the posts not being directly related to fatalism. For instance, one tweet includes a picture of three polar bears sitting on a large pile of plastic and garbage instead of their typical habitat of snow and ice. The text accompanying the picture states that polar bears became scavengers of human food waste, and climate change has impacted polar bears' hunting and forced the animals to move South to locate food. While the post itself does not cover fatalism, for the commenters of the post, it created feelings of hopelessness and anger toward higher organizations that contribute to climate change. Commenters highlighted how we have destroyed the world with greed. One commenter displaying defeatism noted how abnormal, shocking, and *"beyond depressing"* the tweet was. They stated, *"The world is ignorant, and I fear that the planet is doomed. How beautiful and balanced before man came along and unraveled it all."* Another commenter blamed the issue on powerful industries and pointed out that human industrialization is a *"destructive stain on every living thing in nature."* In relation to our survey, the teens felt powerless against climate change and acknowledged the power corporations have over the environment. One participant noted, *"Climate change is the inevitable doom of humanity. Companies won't do anything to change it in fear of losing profits."*

Pessimism. The acknowledgment that animals will be impacted in the future by climate change can create pessimism, and nine posts emphasize animals' bleak future due to climate change. In particular, one Instagram post contains an illustration of a woman with tears in one eye holding an open book. One page of the book says, *"our oceans,"* with fish, coral, and jellyfish, and it is colorful and vibrant. The other page of the book states, *"are dying,"* and there is no more color, and includes dead coral and a plastic bag depicting a frowning face on it. The visual representation portrays a somber outlook on the well-being of animals, particularly aquatic ones, and highlights their worst possible outcomes. While the text of the post relates to an illustration movement to spread environmental awareness and the majority of comments focus on the art itself, one commenter noted how sad and impactful it was to see the

unfortunate future of the animals. Two survey participants directly mentioned how the future of climate change would impact animals negatively. One teen noted how all animals will die due to climate change. Another explained, *"Animals in the Arctic are going to be without a home."*

Shame. The realization of how climate change impacts animals can create shame. For example, one nine-second TikTok video includes images of animals in various states (i.e., healthy versus skinny, covered in plastic, swimming in pollution, and injured; one example includes an injured koala with a cast on its leg) with a song in the background noting, *"You know I can't make it on my own."* The text associated with the video asks to spread awareness and includes hashtags such as #ClimateAction, #SpreadAwareness, and #TakeAction, in addition to #ClimateChange. The overall tone of the post was melancholic, as the video was accompanied by somber background music. Shame was observed in the comment section, with individuals expressing a sense of fear and sadness in regard to the effects of climate change on animals. One commenter mentioned that the post made them feel like screaming and crying, and they asked, *"What has happened to our beautiful planet?"* While the post only had images of animals and no direct discussion of climate change or how the planet is affected, commenters noted how their hearts were broken. Similarly, survey participants explained that thinking about climate change made them sad and scared for their futures: *"I don't feel too well about how I was born into a time where all these climate events are happening, and it's put a damper on how much I normally enjoy life."*

4.2 Hope and Environmental Positivity

Out of all 16 social media posts about animals, 13 of them (81.3%) had a negative tone in both the post and comments. However, two posts that used the #ClimateChange hashtag had a positive perspective on the environment (although not directly related to climate change), and one post expressed hope in the context of animals and climate change. Regarding our survey, a notable disparity was observed between social media posts featuring the #ClimateChange hashtag and how teens discuss climate change online. Specifically, the teens only mentioned engaging with posts containing negative sentiments and did not come across any that included a sense of hope or positivity regarding the environment.

Hope. Many social media posts that feature animals in distress tend to dwell on the negative effects without offering constructive solutions. However, one tweet spotlighted contrasts with the dominant approach. The tweet contains both text and a video and explains, *"Every drop of water has its own story."* In the attached 36-second video, a small bird is on the ground on a road, crying for help and suffering during a heatwave. Someone pours water into a bottle cap and gives it to the suffering bird. The bird struggles to drink the water but is standing up by the end of the video, looking refreshed. The majority of the post comments were positive, and commenters explained how they were pleased with the tweet, saying the video was inspirational. Although commenters still felt bad for the bird, one particular commenter noted that they were happy the tweet was shared and that it was a small step but could

save lives. Encouraging practical and achievable solutions to climate change issues, even small ones, can fill the comments with hopefulness.

Environmental positivity. We found that some social media posts utilizing the #ClimateChange hashtag and highlighting animals do not directly discuss climate change or environmental issues. These posts that focus on the environment in a positive way have the potential to evoke positive emotions related to the environment. For example, on Instagram, there is a post featuring an illustration of a woman and a sloth joyfully riding a bicycle while sitting on top of a whale. The whimsical scene is adorned with bees, jellyfish, fish, flowers, plants, vegetables, and a house. All the creatures, including the woman, are depicted with cheerful smiles. This Instagram post is part of a campaign called “Our Planet Week,” which aims to raise environmental awareness. The post includes an explanation that this illustration movement will contribute to planting a tree for every drawing or illustration created. While the post is tagged #ClimateChange, it mainly focuses on promoting a positive and optimistic perspective about the environment without directly addressing climate change. All comments and reactions to the post are positive, with people saying it puts them in a good mood and that the drawing looks so happy and upbeat. The post brings out positive, cheerful emotions in relation to the environment.

5 DISCUSSION: IMPLICATIONS AND OPPORTUNITIES

We (1) present a way to improve how climate change information is communicated online, especially for teens; (2) contribute implications and open opportunities for attitude/behavior change and activism; and (3) highlight the differences in which #ClimateChange is discussed on Instagram, TikTok, and Twitter in the context of animals and emotion.

Climate change communication and information. For climate change communication and information to have impact online, these discussions need to be about something that teens (or individuals in general) care about. Teens value animals, especially concerning climate change [14, 38]. According to our survey, teens found animal social media posts to be the most impactful and memorable. Wang et al. [46] suggest that in order to engage individuals in climate change discussions, messages need to be relatable and resonate with what matters to them. To help teens care and learn about climate change via social media posts, topics such as animals can be utilized to garner attention and interest. This is important to increase teen engagement with climate change posts, leading to attitude, behavior, and activism opportunities.

Attitudes, behaviors, and activism. To inspire action from social media users on climate change topics, it is crucial to prioritize generating emotionally compelling messages online. Literature highlights how emotions are known to affect attitudes and behaviors [9, 32, 39]. To promote action on climate change, activists may find it effective to appeal to the emotions of their audience. From our content analysis, we saw that the majority of animal-related climate change posts and comments create negative emotions rather than positive ones (and for two positive posts, the #ClimateChange hashtag is used, but the posts do not discuss the crisis explicitly). Further, from our survey, we learned that teens do not have many positive

feelings for climate change posts online. The kind of information we are seeing can lead to climate fatalism rather than positivity. However, we see in the literature that it is beneficial for young people to have positive emotions [30], which can lead to action [48]. In our analysis, we observed one tweet that highlights hope, and hope is a predictor of pro-environmental actions [9]. Further, positive emotions may be used to induce long-lasting, sustainable behavioral changes [9, 10]. Thus, creating positive emotions from climate change discussions online can help teens further develop pro-environmental behaviors, attitudes, and engagement. This is an open opportunity, given that the majority of our data demonstrate that negative emotions are being prompted by the current climate change discourse on these three platforms.

Platform differences. After analyzing the #ClimateChange hashtag on different social media platforms, we found significant differences, including in their portrayal of animals and emotions. TikTok videos showed intense emotional content, highlighting animal harm and negative sentiments. TikTok provides richer media experiences, and we know that teens gravitate toward more interactive content [1]. In contrast, Instagram had a few positive environmental illustrations, evoking feelings of happiness and satisfaction. On Twitter, most tweets about climate change were critical, with only one tweet offering a hopeful perspective centered around animals. Given that we know individuals are operating in a multiple social media platform context, HCI researchers should examine discourse across different platforms [40].

6 LIMITATIONS, CONCLUSION, AND FUTURE WORK

This study has some limitations. We focus on highly engaging and popular social media posts for our content analysis, but this may not provide a comprehensive understanding of #ClimateChange online. Social media posts come in a variety of formats, and the type of posts and media involved (text, images, videos) may have different effects on consumption and engagement. Our research is limited by the small sample size of only 16 animal-related social media posts, which is not representative of all variances on social media. While these considerations exist, this work identifies important opportunities and insights for understanding emotions and discussions about climate change in animal-related social media posts. Given how crucial understanding climate change is for their future and ours, we argue for further developing a research agenda to elucidate teens’ experiences with climate change online, such as assessing the impact of social media activism on teen awareness, knowledge, and understanding of climate change and examining how affordances across platforms connect to climate change engagement. The teens in our survey highlighted that animal-related climate change posts are memorable, and we find value in studying how animals prompt emotional responses and engagement from teens. Researchers can further analyze #ClimateChange and related hashtags to extend the understanding of climate change post characteristics. Additional analysis of comments is crucial as emotions were brought up in the comment sections of posts not originally mentioned in the posts themselves. Our work contributes to a growing research agenda examining how teens interact with climate

change on social media, which can, in turn, support the crucial goal of extending pro-environmental engagement, learning, and action.

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